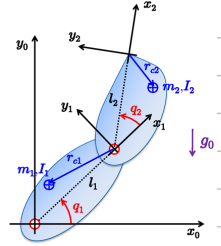
Exercise #1

The 2R robot in Fig. 1 moves in a vertical plane. The two links have, respectively, kinematic lengths l_1 and l_2, masses m_1 and m_2, and barycentric inertias I_1 and I_2 (around the axis normal to the motion plane). The position of the center of mass (CoM) of each link with respect to the attached link frame is given by $\boldsymbol{r}_{ci} = \begin{pmatrix} r_{ci,x} & r_{ci,y} & 0 \end{pmatrix}^T$, with $r_{ci,x} \neq -l_i$ and $r_{ci,y} \neq 0$, for $i = 1, 2$.

A) Determine the robot dynamic model, $\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{c}(\boldsymbol{q}, \dot{\boldsymbol{q}}) + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{u}$, neglecting dissipative effects.

B) Provide a linear parametrization of the model, $\boldsymbol{Y}(\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}})\,\boldsymbol{a} = \boldsymbol{u}$, in terms of a regressor matrix $\boldsymbol{Y} \in \mathbb{R}^{2\times p}$ and a vector $\boldsymbol{a} \in \mathbb{R}^p$ of dynamic coefficients.



The DH transformations are:

$$ {}^0A_1 = \begin{pmatrix} c_1 & -s_1 & 0 & l_1 c_1 \\ s_1 & c_1 & 0 & l_1 s_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad {}^0A_2 = \begin{pmatrix} c_2 & -s_2 & 0 & l_2 c_2 \\ s_2 & c_2 & 0 & l_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \Rightarrow {}^0r_{c1} = {}^0A_1\,{}^1r_{c1} \quad \text{and} \quad {}^0r_{c2} = {}^0A_2\,{}^2r_{c2} $$

We can immidiatly tell that ${}^1\omega_1 = (0 \;\; 0 \;\; \dot{q}_1)^T$ and ${}^2\omega_2 = (0 \;\; 0 \;\; \dot{q}_1 + \dot{q}_2)^T$

$$ {}^0r_{c1} = \begin{pmatrix} c_1 & -s_1 & 0 & l_1 c_1 \\ s_1 & c_1 & 0 & l_1 s_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} r_{c1x} \\ r_{c1y} \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} c_1 r_{c1x} - s_1 r_{c1y} + l_1 c_1 \\ s_1 r_{c1x} + c_1 r_{c1y} + l_1 s_1 \\ 0 \end{pmatrix} $$

$$ {}^0r_{c2} = \begin{pmatrix} c_1 & -s_1 & 0 & l_1 c_1 \\ s_1 & c_1 & 0 & l_1 s_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} c_2 & -s_2 & 0 & l_2 c_2 \\ s_2 & c_2 & 0 & l_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} r_{c2x} \\ r_{c2y} \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} c_{12} & -s_{12} & 0 & l_1 c_1 + l_2 c_{12} \\ s_{12} & c_{12} & 0 & l_1 s_1 + l_2 s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} r_{c2x} \\ r_{c2y} \\ 0 \\ 1 \end{pmatrix} = $$

$$ = \begin{pmatrix} c_{12}(r_{c2x} + l_2) - s_{12} r_{c2y} + c_1 l_1 \\ s_{12}(r_{c2x} + l_2) + c_{12} r_{c2y} + s_1 l_1 \\ 0 \end{pmatrix} $$

$$ v_{c1} = \begin{cases} (-s_1(r_{c1x} + l_1) - c_1 r_{c1y})\dot{q}_1 \\ (c_1(r_{c1x} + l_1) - s_1 r_{c1y})\dot{q}_1 \end{cases} \quad v_{c2} = \begin{cases} (\dot{q}_1 + \dot{q}_2)(-s_{12}\overbrace{(r_{c2x} + l_2)}^{R_2} - c_{12} r_{c2y}) - s_1 l_1 \dot{q}_1 \\ (\dot{q}_1 + \dot{q}_2)(c_{12}(r_{c2x} + l_2) - s_{12} r_{c2y}) + c_1 l_1 \dot{q}_1 \end{cases} $$

$$ T_1 = \frac{1}{2} m_1 \dot{q}_1^2 \left[\left(-s_1 \overbrace{(r_{c1x} + l_1)}^{R_1} - c_1 r_{c1y}\right)^2 + \left(c_1(r_{c1x} + l_1) - s_1 r_{c1y}\right)^2 \right] + \frac{1}{2} I_1 \dot{q}_1 = $$

$$ \frac{1}{2}\dot{q}_1^2 \left[m_1 \left(s_1^2 R_1^2 + c_1^2 r_{c1y}^2 + 2 c_1 s_1 R r_{c1y} + c_1^2 R^2 + s_1^2 r_{c1y}^2 - 2 c_1 s_1 R r_{c1y} \right) + I_1 \right] = $$

$$ \frac{1}{2}\dot{q}_1^2 \left[m_1 (R_1^2 + r_{c1y}^2) + I_1 \right] $$

$$ \begin{aligned} T_2 = \frac{1}{2} m_2 \Big[& (\dot{q}_1^2 + \dot{q}_2^2 + \dot{q}_1\dot{q}_2)(s_{12}^2 R_2^2 + c_{12}^2 r_{c2}^2 + 2 c_{12} s_{12} R_2 r_{c2y}) + \dot{q}_1^2 l_1^2 s_1^2 \\ & + 2(\dot{q}_1^2 + \dot{q}_1\dot{q}_2)(s_{12} s_1 R_2 l_1 + c_{12} s_1 r_{c2y} l_1) \\ & (\dot{q}_1^2 + \dot{q}_2^2 + \dot{q}_1\dot{q}_2)(c_{12}^2 R_2^2 + s_{12}^2 r_{c2y}^2 - 2 c_{12} s_{12} R_2 r_{c2y}) + \dot{q}_1^2 l_1^2 c_1^2 \\ & + 2(\dot{q}_1^2 + \dot{q}_1\dot{q}_2)(c_{12} c_1 R_2 l_1 - s_{12} c_1 r_{c2y} l_1) \Big] + \frac{1}{2} I_2 (\dot{q}_1^2 + \dot{q}_2^2 + \dot{q}_1\dot{q}_2) \end{aligned} $$

$$ \begin{aligned} T_2 = \frac{1}{2} m_2 \Big[& (\dot{q}_1^2 + \dot{q}_2^2 + \dot{q}_1\dot{q}_2)(R_2^2 + r_{c2y}^2) + \dot{q}_1^2 l_1^2 + 2(\dot{q}_1^2 + \dot{q}_1\dot{q}_2)(R_2 l_1 c_2 - r_{c2y} l_1 s_2) \Big] \\ & + \frac{1}{2} I_2 (\dot{q}_1^2 + \dot{q}_2^2 + \dot{q}_1\dot{q}_2) \end{aligned} $$

$$T = \frac{1}{2}\dot{q}_1^2\left[m_1(R_1^2+r_{c1y}^2) + I_1 + I_2 + m_2\left(l_1^2 + (R_2^2+r_{c2y}^2) + 2(\underbrace{R_2 l_1 c_2 - r_{c2y} l_1 s_2}_{= \rho_3 + 2\rho_1 c_2 + 2\rho_2 s_2})\right)\right]$$

$$+ \frac{1}{2}\dot{q}_2^2\left[m_2(R_2^2+r_{c2y}^2) + I_2\right] = \rho_4$$

$$+ \frac{1}{2}\dot{q}_1\dot{q}_2\left[m_2\left(2(R_2^2+r_{c2y}^2) + 2(R_2 l_1 c_2 - r_{c2y} l_1 s_2) + I_2\right]\right. = \rho_4 + \rho_1 c_2 + \rho_2 s_2$$

$$M(q) = \begin{pmatrix} 2(\rho_1 c_2 + \rho_2 s_2) + \rho_3 & \rho_4 + \rho_1 c_2 + \rho_2 s_2 \\ \rho_4 + \rho_1 c_2 + \rho_2 s_2 & \rho_4 \end{pmatrix}$$

At this point I compute $C(q, \dot{q})$

$$C_1 = \frac{1}{2}\left(\frac{\partial M_1}{\partial q} + \left(\frac{\partial M_1}{\partial q}\right)^T - \frac{\partial M}{\partial q_1}\right) = \begin{pmatrix} 0 & \rho_2 c_2 - \rho_1 s_2 \\ \rho_2 c_2 - \rho_1 s_2 & \rho_2 c_2 - \rho_1 s_2 \end{pmatrix}$$

$$C_2 = \frac{1}{2}\left(\frac{\partial M_2}{\partial q} + \left(\frac{\partial M_2}{\partial q}\right)^T - \frac{\partial M}{\partial q_2}\right) = \begin{pmatrix} \rho_1 s_2 - \rho_2 c_2 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\begin{cases} c_1 = \dot{q}^T C_1 \dot{q} = (2\dot{q}_1\dot{q}_2 + \dot{q}_2^2)(\rho_2 c_2 - \rho_1 s_2) \\ c_2 = \dot{q}_1^2(\rho_1 s_2 - \rho_2 c_2) \end{cases}$$

The last term to compute is $g(q)$.

$$U = g_0\left(m_1(s_1(r_{c1x} + l_1) + c_1 r_{c1y}) + m_2(s_{12}(r_{c2x} + l_2) + c_{12} r_{c2y} + s_1 l_1)\right)$$

$$g_1 = \frac{\partial U}{\partial q_1} = g_0\left(m_1(c_1(r_{c1x} + l_1) - s_1 r_{c1y}) + m_2(c_{12}(r_{c2x} + l_2) - s_{12} r_{c2y} + c_1 l_1)\right)$$

$$g_2 = \frac{\partial U}{\partial q_2} = g_0 m_2\left(c_{12}(r_{c2x} + l_2) - s_{12} r_{c2y} + c_1 l_1\right)$$ model complete.

Exercise #2

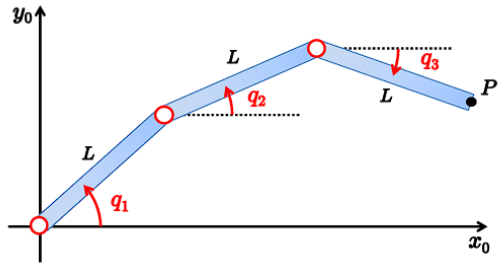
Consider the planar 3R robot with links of unitary length in Fig. 2 and use the absolute coordinates $\boldsymbol{q} = (q_1, q_2, q_3)$ defined therein. The robot is commanded with the joint velocity $\dot{\boldsymbol{q}} \in \mathbb{R}^3$. Starting from the configuration $\boldsymbol{q}(0) = (\pi/4, 0, 0)$, the robot should simultaneously move its end-effector point P along a vertical line parallel to $\boldsymbol{y}_0$ with a constant speed $v > 0$, while keeping the second link horizontal. Determine the first encountered configuration $\boldsymbol{q}_s$ at which these two tasks run into an algorithmic singularity. In $\boldsymbol{q} = \boldsymbol{q}_s$ and for $v = 1$ [m/s], compute the following three commands:

a) $\dot{\boldsymbol{q}}_{PS}$ using pseudoinversion of the extended Jacobian of the two tasks;

b) $\dot{\boldsymbol{q}}_{DLS}$ using damped least squares on the extended Jacobian, with damping parameter $\mu^2 = 0.25$;

c) $\dot{\boldsymbol{q}}_{TP}$ using the task priority method, with the end-effector task having the highest priority.

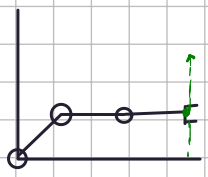
Compare in the three cases the norm of the resulting joint velocity, the norm of the end-effector velocity error $\dot{\boldsymbol{e}}_P \in \mathbb{R}^2$, and the absolute value of the second joint velocity error $\dot{e}_{q_2} \in \mathbb{R}$.



the task vector concerns $P \in \mathbb{R}^2$ and q_2 so:

$$DK: \mathcal{E}(q) = \begin{cases} c_1 + c_2 + c_3 \\ s_1 + s_2 + s_3 \\ q_2 \end{cases} \Rightarrow J = \begin{pmatrix} -s_1 & -s_2 & -s_3 \\ c_1 & c_2 & c_3 \\ 0 & 1 & 0 \end{pmatrix}$$

the desired $\dot{\mathcal{E}}$ is $\dot{\mathcal{E}}_d = (0 \quad v \quad 0)^T$, starting from $q(0) = (\frac{\pi}{4}, 0, 0)^T$



Since $q_2 = 0 \Rightarrow$ $J = \begin{pmatrix} -s_1 & 0 & -s_3 \\ c_1 & 1 & c_3 \\ 0 & 1 & 0 \end{pmatrix}$ $\Rightarrow$ this became singular in the exact moment $q_1 = q_3$

So in that configuration $q_s = (\alpha, 0, \alpha)^T$

$$\mathcal{E}(q) = \begin{cases} 1 + 2\cos\alpha = q_1(0) = 2.707 \\ 2\sin\alpha = \beta \\ 0 \end{cases} \Rightarrow \begin{cases} \cos\alpha = \frac{1}{2} 1.707 \\ \sin\alpha = \beta/2 \end{cases} \Rightarrow \begin{cases} \alpha = q_1 = q_3 = 0.548 \\ \beta = 1.0421 \end{cases}$$

$q_s \simeq (0.548, 0, 0.548)^T$

in that conf. the Jacobian is

$$J = \begin{pmatrix} -0.52 & 0 & -0.52 \\ 0.853 & 1 & 0.853 \\ 0 & 1 & 0 \end{pmatrix} \quad \text{and} \quad \dot{\mathcal{E}}_d = (0 \quad 1 \quad 0)^T$$

ovviamente i calcoli si fanno con Matlab, anche perché la pseudo inversa di J richiede la scomposizione ai valori singolari.

$$\dot{q}_{PS} = J^{\#}(0 \ 1 \ 0)^T = (0.3362, 0.2132, 0.3362)^T$$

in the DLS method:

$$\dot{q}_{DLS} = J^T(JJ^T + \tfrac{1}{4} I_m)^{-1} \dot{\mathcal{E}}_d = (0.2963 \quad 0.2198 \quad 0.2963)^T$$

With task priority:

$$P_2 = I - J_2^{\#} J_2$$

$$\dot{\mathcal{E}}_1 = (0 \ 1)^T \qquad \dot{\mathcal{E}}_2 = 0 \qquad J_1 = \begin{pmatrix} -0.52 & 0 & -0.52 \\ 0.853 & 1 & 0.853 \end{pmatrix} \qquad J_2 = (0 \ 1 \ 0)$$

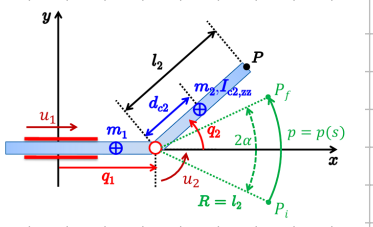
$$\dot{q}_1 = J_1^{\#} \dot{\mathcal{E}}_1 = (0 \ 1 \ 0) \qquad \dot{q}_{TP} = \dot{q}_2 = \dot{q}_1 + (J_2 P_2)^{\#}(\dot{\mathcal{E}}_2 - J_2 \dot{q}_1) = (0 \quad 1 \quad 0)^T$$

Exercise #3

The dynamic model of a PR robot moving on a horizontal plane is given in the lecture slides[1]. The end-effector point P should trace in minimum time a circular path of radius $R = l_2$

$$\boldsymbol{p}(s) = \begin{pmatrix} k + R\cos(s-\alpha) \\ R\sin(s-\alpha) \end{pmatrix}, \qquad s = [0, 2\alpha] \qquad (0 < \alpha < \frac{\pi}{2}),$$

from rest to rest between P_i and P_f (see Fig. 3), under bounded input force/torque $|u_i| \le U_{i,max}$, $i = 1, 2$. Assuming that the second bound $U_{2,max}$ is the only limiting factor, provide the expression of the needed input $\boldsymbol{u}_d(t) \in \mathbb{R}^2$ and of the minimum time T. Sketch the time profiles of the inputs.



the DK is: $\begin{cases} q_1 + l_2 c_2 = k + R\cos(s-\alpha) \\ l_2 s_2 = R\sin(s-\alpha) \end{cases} \Rightarrow q(s) = \begin{cases} k \\ s-\alpha \end{cases} \quad \dot{q} = \begin{cases} 0 \\ \dot{s} \end{cases} \quad \ddot{q} = \begin{cases} 0 \\ \ddot{s} \end{cases}$

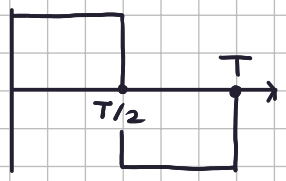
Since is rest to rest, $s(t)$ should be a bang-coast-bang.

$$-m_2 d_{c2}\sin(s-\alpha)\ddot{s} - m_2 d_{c2}\cos(s-\alpha)\dot{s}^2 = u_1$$

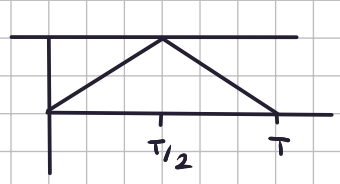
$$(m_2 d_{c2}^2 + I_{c2zz})\ddot{s} = u_2, \quad |u_2| \le U_2$$

$$\Rightarrow \quad |\underbrace{(m_2 d_{c2}^2 + I_{c2zz})}_{\rho}\ddot{s}| \le U_2 \Rightarrow |\ddot{s}| \le \frac{1}{\rho}U_2 \Rightarrow \text{the max } \ddot{s} \text{ is } \frac{1}{\rho}U$$

$s(t)$ should be a b-c-b.

$\ddot{s}(t) =$ 

$\Rightarrow v_{max} = \frac{T}{2}\cdot\frac{1}{\rho}U$

$\dot{s}(t) =$ 

$$\Rightarrow \underbrace{T\cdot\frac{T}{2\rho}U\frac{1}{2}}_{\text{total displacement}} = 2\alpha \Rightarrow T = \left(\frac{1}{U_{2max}}8\alpha(m_2 d_{c2}^2 + I_{c2zz})\right)^{1/2}$$